

Multi-trauma: Blunt VSA + Cricothyrotomy



Section I: Scenario Demographics

Scenario Title:	Multi-trauma MVC - Cricothyrotomy + Blunt VSA		
Date of Development:	27/04/206 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups		

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a multi-trauma where they must manage two critically ill patients at once.
CRM Objectives:	1) Demonstrates leadership by maintaining control during crisis 2) Triage patients, prioritizes tasks, and appropriately allocates resources dedicated to each patient (termination of care in blunt trauma VSA patient)
Medical Objectives:	1) Recognizes the potential for a difficult airway and plans accordingly 2) Identifies the need for a surgical airway and performs a surgical cricothyrotomy

Case Summary: Brief Summary of Case Progression and Major Events
The case will start with an EMS patch indicating that they are 2 minutes out with multi-trauma from a 2 car MVC. Two patients will then arrive within 1 minute of each other. The first will have gone VSA en route from presumed blunt trauma. This patient will not regain a pulse. The second patient will arrive with significant burns from a car fire, and will have GCS of 3 necessitating intubation. All attempts at intubation will be unsuccessful, and a surgical airway must be performed. The team will need to prioritize resources between the two patients and realize that an ED thoracotomy is not reasonable in the first patient.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin ☐ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☐ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates	Brief Description of Role		
Paramedics	To bring the two patients and describe their management.		
Nurses	One per patient to assist at bedside, cue team to patient features		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☒ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☒ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask on both patients at start of case	☒ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☒ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Patient 1: Hematoma to occiput, bilateral bruising over anterior chest, bruising over abdominal flanks. Splinted R arm. Pelvic binder. C-collar Patient 2: Singed eyebrows and soot on face. Burns to L arm and L leg with some burns on left side of chest. C-collar			
E. Approximate Timing			
Set-Up:	15 min	Scenario:	20 min
		Debriefing:	40 min

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Section V: Patient Data and Baseline State (PATIENT #1)

A. Clinical Vignette: To Read Aloud at Beginning of Case

You are working in a tertiary care trauma center. *EMS patch: We have a 50ish M unbelted driver in a head-on MCV at about 60km/hr. He was ejected from the vehicle and found about 30m from the crash site with a GCS of 3. He has an obvious head injury, torso injury and unstable pelvis, which we've bound. Initially had RR 40, O2 85% on NRB, HR 150 and a questionable femoral pulse. Since then, he's been pulseless. We've been en route about 5 minutes and should be there in about 2 min. He's received 1mg Epi so far with no shocks advised x2. Smells of EtOH, but no other known history. There was one other car involved that caught on fire, so you'll probably get them, too, if they survive. Please prepare for this patient.*

B. Patient Profile and History

Patient Name: John Doe		Age: 56		Weight: 100kg	
Gender: ☒ M ☐ F		Code Status: Full			
Chief Complaint: MCV → VSA					
History of Presenting Illness: As above.					
Past Medical History:	unknown	Medications:	unknown		
Allergies: unknown					
Social History: unknown					
Family History: unknown					
Review of Systems:	CNS:	Unable			
	HEENT:	Unable			
	CVS:	Unable			
	RESP:	Unable. Breath smells of EtOH			
	GI:	Unable			
	GU:	Unable			
	MSK:	Unable	INT:	Unable	

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 0	BP: 0	RR: 14/min (bagged)	O ₂ SAT: 85 % NRB		
Rhythm: asystole	T: 35°C	Glucose: 7 mmol/L	GCS: 3 (E V M)		
General Status: GCS3. Active CPR ongoing by EMS.					
CNS:	No movement. No gag to oral airway.				
HEENT:	Large boggy hematoma over occiput. Pupil 6mm on R fixed, 3mm reactive on L. C-collared				
CVS:	Normal S1, S2. No murmurs.				
RESP:	Decreased air entry bilaterally. Bilateral crepitus. Bruising/abrasions to chest.				
ABDO:	Distended. Bruising over flanks.				
GU:	No visible blood at urethral meatus.				
MSK:	Pelvis bound. Deformed R arm splinted.		SKIN:	Mottled.	

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Section VI: Scenario Progression (PATIENT #1)

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Preparation		☐ Assigns tasks to team members prior to patient arrival ☐ Prepares equipment	
2. Baseline State Rhythm: asystole HR: 0 BP: 0 RR: 14/min (bagged) O ₂ SAT: 75 % NRB T: 35°C	Unresponsive. No gag with oral airway.	<u>Learner Actions</u> ☐ IV, O ₂ , monitors ☐ ± Continue CPR ☐ Cardiac POCUS to assess for cardiac activity: reasonable to call code if none ☐ Level 1 transfuser ☐ Call for pRBC (O+), transfuse 2 units rapidly ☐ Bilateral finger thoracostomy ± chest tube insertion ☐ Check pelvic binder placement ☐ Intubation ☐ ± Epi 1mg IV q2-3 min ☐ ± FAST exam ☐ ± TxA 1g IV over 10 mins ☐ Consider calling code	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - 2 minutes or attempt thoracostomy → 2nd patient arrives - Epi/CPR → no ROSC - Needle decompression → air - Finger thoracostomy → air and blood back bilaterally - Chest tube insertion → 500cc on L and 600cc on R, O ₂ 82% - Intubation → O ₂ 84% <u>Triggers</u> <i>For progression to next state</i> - Team leader decides to end resuscitation → END CASE

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Section V: Patient Data and Baseline State (PATIENT #2)

A. Clinical Vignette: To Read Aloud at Beginning of Case

EMS Handover: This 30ish male belted driver was in a head on MVC with both cars going ~60km/hr. His car was on fire when we got there, and he's got 2nd/3rd degree burns everywhere. We found him outside the car, so he must have self-extricated. His GCS has been 3 the entire time with us. He's tolerating an oral airway. His last vitals were HR 120, BP 130/80, RR 30, O2 95% NRB. (If asked: cap sugar 6.3)

B. Patient Profile and History

Patient Name: Shawn Sherman	Age: 33	Weight: 70
Gender: ☒ M ☐ F	Code Status: Full	
Chief Complaint: burns, aLOC		
History of Presenting Illness: As above.		
Past Medical History:	Unknown	Medications: Unknown
Allergies: None known.		
Social History: Unknown		
Family History: Unknown.		
Review of Systems:	CNS: Unable	
	HEENT: Unable	
	CVS: Unable	
	RESP: Unable	
	GI: Unable	
	GU: Unable	
	MSK: Unable	INT: Unable

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 130/min		BP: 135/85		RR: 30/min	
Rhythm: sinus tach		T: 36.1°C		Glucose: 6.1 mmol/L	
				O ₂ SAT: 95% NRB	
				GCS: 3 (E 1 V 1 M1)	
General Status: Unresponsive.					
CNS:	GCS 3. Pupils 3mm bilat, minimally reactive.				
HEENT:	No signs HI. Soot on face and in nares. Singed eyebrows.				
CVS:	Nil.				
RESP:	GAEB. No adventitious.				
ABDO:	Abdo soft, NT.				
GU:	Nil.				
MSK:	No signs trauma. Pelvis stable.			SKIN:	Burns to part of left chest and along left arm and leg. (TBSA = approx. 25%)

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Section VI: Scenario Progression (PATIENT #2)

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 130/min BP: 90/70 RR: 30/min O ₂ SAT: 96 % NRB T: 36.1°C	Unresponsive with GCS 3.	Learner Actions ☐ 2 large bore IVs, bolus 1L ☐ 100% O₂, monitors ☐ Labs: VBG, carboxyHb, lactate, coags, trop, G&S, INR ☐ Check glucose: 6.3 ☐ Portable CXR/PXR ☐ Full exposure re: TBSA ☐ US FAST exam (negative)	Modifiers <i>Changes to patient condition based on learner action</i> - 1L IVF → HR 120, BP 120/80 Triggers <i>For progression to next state</i> - Intubate → 2. Intubation
2. Intubation Rhythm: sinus HR: 120 BP: 90/70 RR: 30/min O ₂ SAT: 93 % NRB	Unchanged.	Learner Actions ☐ Push dose pressors at bedside ☐ Consider apneic oxygenation ☐ Difficult airway cart ☐ Surgical airway kit at bedside and neck prepped ☐ In-line stabilization ☐ Low dose sedation ± paralytic ☐ Anesthesia consult ☐ Surgical cricothyroidotomy (bougie cric ideal, any method acceptable)	Modifiers - Propofol used → BP 60/30 - Paralytic/sedative given → O₂ decreases to 75% over 3 minutes (no improvement with BMV) - Intubation without paralytic or sedative → O₂ decreases to 85% over 3 minutes (BMV no help) - All intubation attempts unsuccessful → "very edematous, cannot see any landmarks" Triggers - No SCT by 10 minutes → 3. Hypoxic Arrest - SCT and airway secure → 4. Resolution
3. Hypoxic Arrest Rhythm → PEA BP → -/-	Patient pulseless.	Learner Actions ☐ Ensure quality CPR ☐ Epinephrine q3min ☐ HCO₃ ampules ☐ Surgical Cricothyrotomy (bougie cric ideal, any method acceptable)	Modifiers Triggers - SCT and airway secure → 4. Resolution
4. Resolution Rhythm: sinus HR: 110 BP: 100/50 RR: 12/min vented O ₂ SAT: 95%	Patient remains unresponsive.	Learner Actions ☐ Call ICU ☐ Call Plastics/Burn Centre ☐ Complete secondary survey ☐ Repeat CXR ☐ Arrange for CT	END CASE PRN



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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results – NONE given for either patient

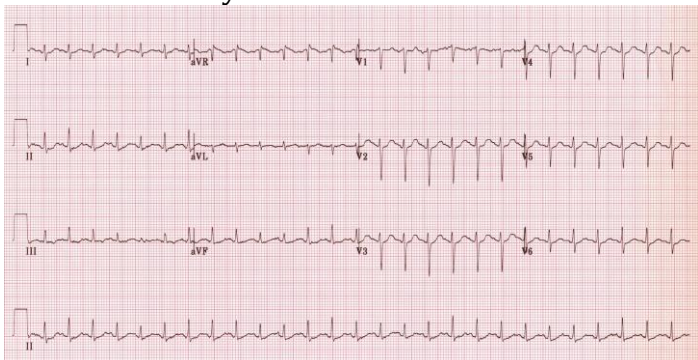
Images – PATIENT 1 (Ultrasound only)

Cardiac U/S – cardiac standstill

FAST – FF in RUQ

Images (ECGs, CXRs, etc.) – PATIENT #2

ECG: Sinus tachycardia



(ECG source: <https://lifeinthefastlane.com/ecg-library/sinus-tachycardia/>)

CXR: normal pre-intubation



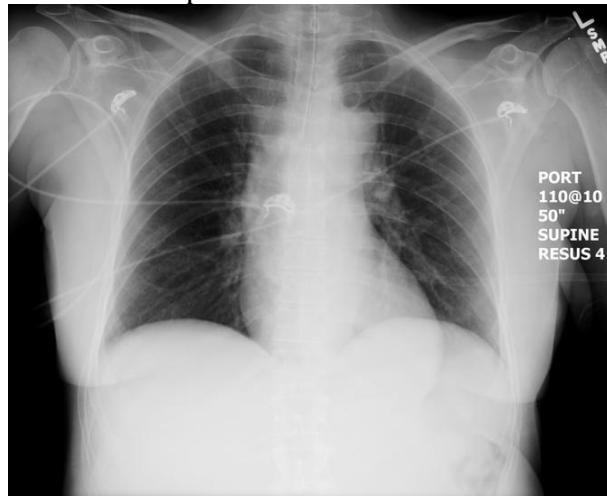
(CXR source: <https://radiopaedia.org/cases/normal-chest-x-ray>)

PXR: Normal



(PXR source: <http://radiopaedia.org/articles/pelvis-1>)

CXR: normal post-intubation



(CXR source: <https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg>)

Ultrasound Video Files (if applicable)

No FF in abdomen

No PTX bilaterally (initial and if repeat)

No PCE

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To expose learners to a multi-trauma where they must manage two critically ill patients at once.
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Medical Objectives:	1) Recognizes the potential for a difficult airway and plans accordingly 2) Identifies the need for a surgical airway and performs a surgical cricothyrotomy
Sample Questions for Debriefing	
1) What was your approach to resource utilization in this case? How did your care of patient 1 change, if at all? 2) What was your approach to intubation in the patient in the car fire? What would you do differently next time? What do you think you did well? When did you realize the need for SCT? 3) What was your approach to fluid resuscitation in the second burn patient? What is the utility of the Parkland formula in this case, if any?	
Key Moments	
1) Decision to terminate resuscitation in the VSA blunt trauma patient and reallocate resources	
2) Anticipating a difficult airway in the burn patient	
3) Recognizing the need for SCT	